Problem 4: Evaluation and Interpretation

Git repository: https://github.com/mfarmani95/Streamflow_Prediction.git

Methods: I evaluated the best LSTM and Transformer runs on held-out test basins. Both models use meteorological forcing sequences and static catchment attributes. The split is by basin, avoiding basin leakage between training, validation, and test.

LSTM run: outputs/sweeps_mse/seq120_hidden128_batch032_lr0p001

LSTM configuration: model=lstm, loss=nse, seq_len=120, window_stride=120, hidden_size=128, num_layers=1, dropout=0.1, batch_size=32, lr=0.001, train_basin_count=30, val_basin_count=10, test_basin_count=10, split_strategy=stratified

Transformer run: output/transformer_MSE_T1/seq120_hidden064_batch128_lr0p001_drop0p2

Transformer configuration: model=transformer, loss=mse, seq_len=120, window_stride=120, hidden_size=64, num_layers=2, nhead=4, dim_feedforward=128, dropout=0.2, batch_size=128, lr=0.001, train_basin_count=30, val_basin_count=10, test_basin_count=10, split_strategy=stratified

Because window_stride equals seq_len, the plotted hydrographs show non-overlapping sequence-target predictions rather than rolling daily predictions.

Overall Held-Out Test Metrics

Model	NSE	KGE	RMSE	MAE	MSE
LSTM	0.5238	0.7593	2.1738	0.9174	4.7254
Transformer	0.2765	0.6265	2.6794	1.0624	7.1789

Catchment KGE CDF Summary

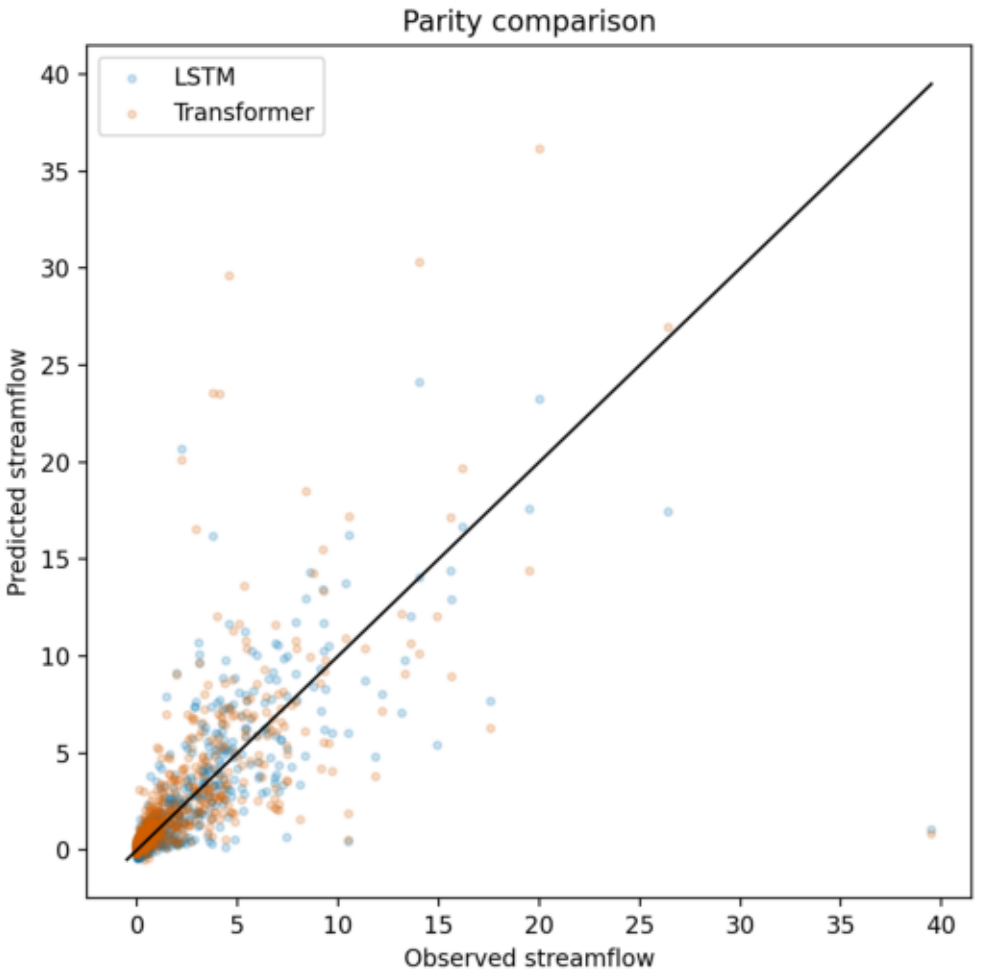
Model	Median KGE	KGE < 0	KGE >= 0.5	KGE >= 0.7
LSTM	0.5448	30.0%	50.0%	30.0%
Transformer	0.2889	40.0%	40.0%	20.0%

Low-Flow and Peak-Flow Metrics

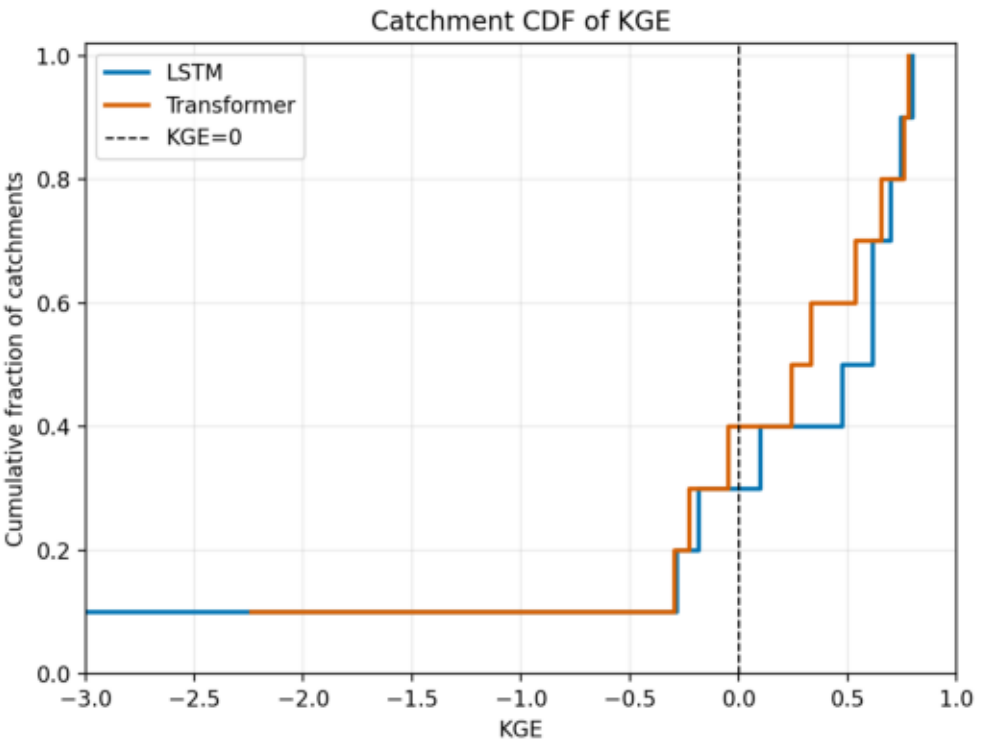
Model	Regime	NSE	Percent bias	n
LSTM	low_flow_obs_le_p10	-9414.0923	14885.0%	88.0000
LSTM	peak_flow_obs_ge_p95	-0.6044	-17.2%	44.0000
Transformer	low_flow_obs_le_p10	-7940.1909	1102.5%	88.0000
Transformer	peak_flow_obs_ge_p95	-0.9949	-13.8%	44.0000

Scatter and Catchment Skill

Predicted vs. observed

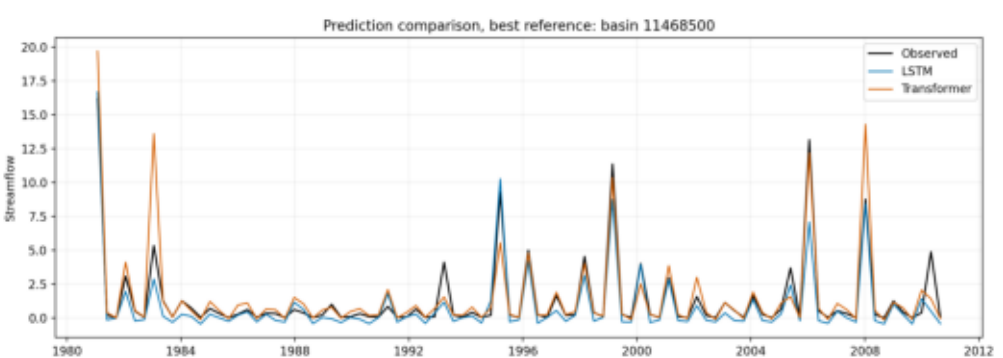


KGE CDF over catchments

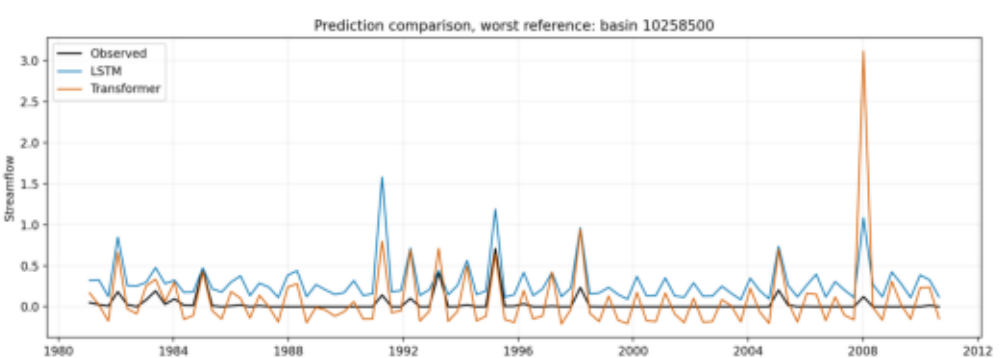


Best and Poorly Performing Basins

Best reference basin

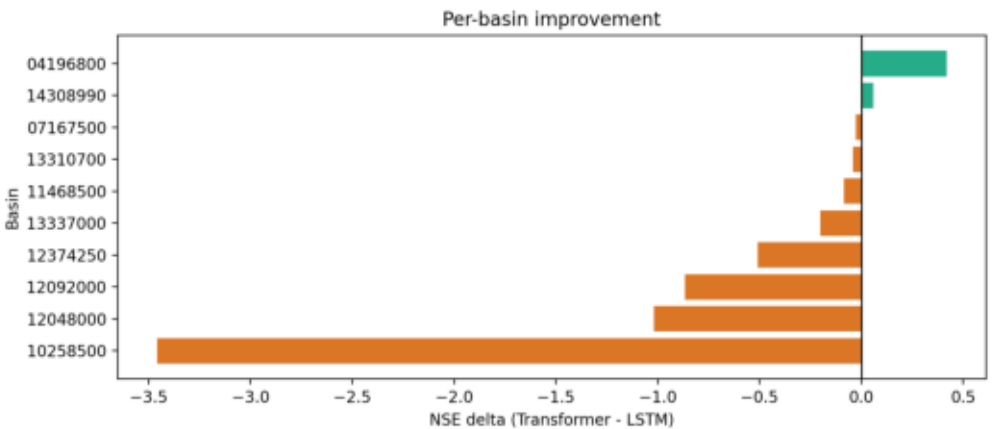


Worst reference basin

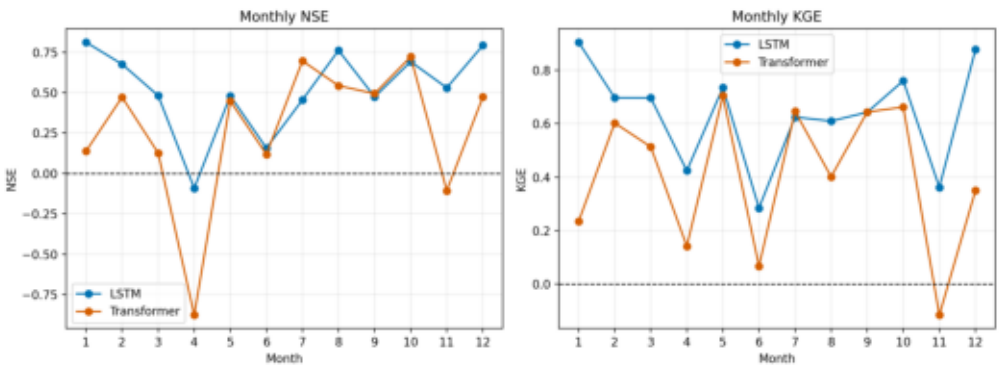


Basin and Seasonal Diagnostics

Per-basin NSE delta

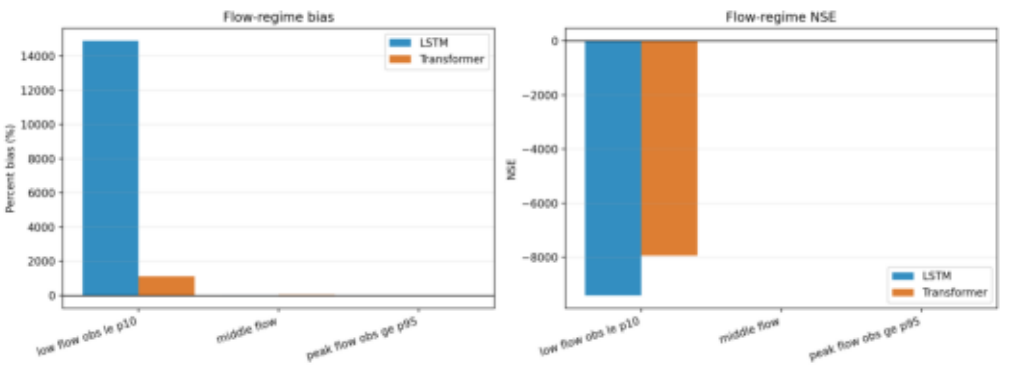


Monthly NSE/KGE

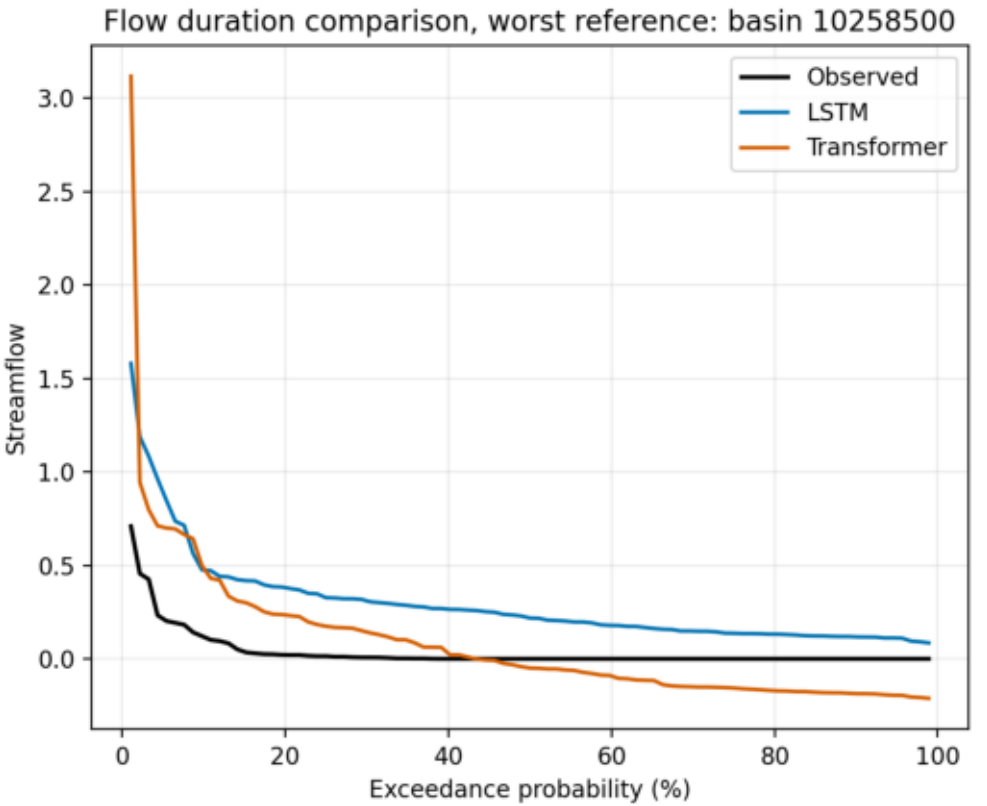


Hydrologic Regime Diagnostics

Low/middle/peak flow behavior



Worst-basin flow duration curve



Interpretation and Conclusion

For LSTM, the best basin by NSE was 11468500 (NSE=0.862, KGE=0.616, RMSE=1.062). The poorest basin was 10258500 (NSE=-8.456, KGE=-6.240, RMSE=0.324).

For Transformer, the best basin by NSE was 14308990 (NSE=0.910, KGE=0.761, RMSE=0.677). The poorest basin was 10258500 (NSE=-11.915, KGE=-2.238, RMSE=0.378).

Relative to the LSTM, the Transformer had lower overall NSE (delta=-0.247), KGE delta=-0.133, and RMSE delta=0.506. The basin-level comparison is therefore important: overall means can hide whether one model improves many catchments or only a few difficult ones.

LSTM catchment KGE distribution had median KGE=0.545; 30.0% of catchments were below KGE=0, 50.0% were at or above KGE=0.5, and 30.0% were at or above KGE=0.7.

Transformer catchment KGE distribution had median KGE=0.289; 40.0% of catchments were below KGE=0, 40.0% were at or above KGE=0.5, and 20.0% were at or above KGE=0.7.

LSTM had its strongest monthly NSE in month 1 (NSE=0.812) and weakest monthly NSE in month 4 (NSE=-0.092).

Transformer had its strongest monthly NSE in month 10 (NSE=0.722) and weakest monthly NSE in month 4 (NSE=-0.876).

LSTM flow-regime behavior: low flow obs le p10 bias=14885.0% and NSE=-9414.092; middle flow bias=10.2% and NSE=0.225; peak flow obs ge p95 bias=-17.2% and NSE=-0.604.

Transformer flow-regime behavior: low flow obs le p10 bias=1102.5% and NSE=-7940.191; middle flow bias=31.6% and NSE=-0.473; peak flow obs ge p95 bias=-13.8% and NSE=-0.995.

The LSTM provides a recurrent memory structure, while the Transformer can attend across the sequence more directly. The comparison should be interpreted at both overall and basin levels because a model can improve the mean score while still failing in specific catchments or flow regimes.

The main successful part of the workflow is useful streamflow skill in some basins and a clear basin-held-out evaluation. The main weaknesses are uneven basin performance, low-flow bias, and peak-flow underestimation. Future work should test rolling predictions, extreme-flow-aware objectives, and additional catchment/process information.